T-1 (Tutorial / Recitation) - 1

9/01/2026

1. A smart farming system predicts whether a plant will grow well (1) or poorly (0) based on daily sunlight hours and water amount in liters.

Dataset:

Sunlight Hours (x_1)	Water Liters (x_2)	Bias (x_3)	Target (Good growth = 1 / Poor)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1

Sol:- Inputs (x_1 & x_2) Output (Target)

- Mean / Average : Central tendency of the data
- Variance / standard Deviation :

- Mean of x_1 $= \frac{3+8+5+10+7}{5} = \frac{33}{5} = 6.6 = \mu_{x_1}$

- Mean of x_2 $= \frac{1+4+2+5+3}{5} = \frac{15}{5} = 3 = \mu_{x_2}$

* Variance, $\boxed{(\sigma^2) = \frac{1}{N} \sum_i (x_i - \mu)^2}$

N → no. of data.

μ → input Mean

$$= \frac{(3-6.6)^2 + (8-6.6)^2 + (5-6.6)^2 + (10-6.6)^2 + (7-6.6)^2}{5}$$

$$= \frac{146}{25} = 5.84 =$$

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$$= \frac{(1-3)^2 + (4-3)^2 + (2-3)^2 + (5-3)^2 + (3-3)^2}{5}$$

$$= 2.$$

$\Rightarrow$ Standard Deviation $= \sqrt{\sigma^2_{x_1}} = \sqrt{(5.8)} = 2.40$

$$= \sqrt{\sigma^2_{x_2}} = \sqrt{2} = 1.414.$$

ML Model $\rightarrow$ Perception Model.

$\Rightarrow$ Simplest Neutral Network (N.N) model.

$\Downarrow$

Simple decision making model that can Answer Yes/No

$\Rightarrow$ Preception:

1) Inputs :- x_1 and x_2
2) Weights associated with inputs: w_1 and w_2
3) Bias $\rightarrow$ Model flexiability (b)
4) Weighted sum: $z = w_1x_1 + w_2x_2 + b.$

$$= \boxed{\sum_i w_i x_i + b}$$

5) Activation function: $\hat{y} = \begin{cases} 1; & \text{if } z \geq 0; \\ 0; & \text{if } z < 0 \end{cases}$
(Step function)

6) Updated weights:

$$w_i^{New} = \boxed{w_i^{old} + \eta(y - \hat{y}) \times x_i}$$

$\eta \rightarrow$ learning Rate

$y \rightarrow$ Actual output

$\hat{y} \rightarrow$ Predicted Target

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Prediction

1) Initial:- Inputs : $x_1 = 3$; $x_2 = 1$; bias = 1

let, $w_1 = 0$

$w_2 = 0$.

$\eta = 1$

$y = 0$.

$z = ?$

$z = w_1 x_2 + w_2 x_2 + b$

$= 0 \times 3 + 0 \times 1 + 1$

$= 1$.

$z = 1$; $\hat{y} = 1$

$w_1^{New} = 0 + 1(0 - 1) \times 3 = -3$

$w_2^{New} = 0 + 1(0 - 1) \times 1 = -1$

Now, $w_1 = -3$

$w_2 = -1$

$\eta = 1$

$b = 1$

Inputs: $x_1 = 8$; $x_2 = 4$; $b = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$= -3 \times 8 + -1 \times 4 + 1$

$= -24 - 4 + 1$

$= -27$;

$z = -27$; $\hat{y} = 0$

$y = 1$.

$w_1^{New} = -3 + 1(1 - 0) \times 8 = -2 \times 8 = -5$

$w_2^{New} = -1 + 1(1 - 0) \cdot 4 = = 0 = 3$.

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Now, $w_1 = 5$: Inputs: $x_1 = 5$; $x_2 = 2$; $b = 1$

$w_2 = 3$ $y = 0$.

$\eta = 1$ $z = 5 \times 5 + 3 \times 2 + 1$ $w_i^{New} = w_i^{old} + \eta(y - \hat{y}) x_i$.

$b = 1$ $= 25 + 6 + 1$

$z = 32$; $\hat{y} = 1$.

$w_1^{New} = 5 + 1(0 - 1) \times 5 = 0$

$w_2^{New} = 3 + 1(0 - 1) \times 2 = 1$.

Now, $w_1 = 0$. : Inputs: $x_1 = 0$; $x_2 = 5$; $b = 1$.

$w_2 = 1$. $y = 1$.

$\eta = 1$

$b = 1$. $z = 10 \times 0 + 1 \times 5 + 1$

$= 5$.

$z = 5$; $5 \geq 0$. $\hat{y} = 1$.

$w_1^{New} = 0 + 1(1 - 1) \times 10 = 0$

$w_2^{old} = 1 + 1(1 - 1) \times 5 = 1$.

Now, $w_1 = 0$. : Inputs: $x_1 = 7$, $x_2 = 3$; $b = 1$

$w_2 = 1$. $y = 1$.

$\eta = 1$

$b = 1$. $z = 7 \times 0 + 3 \times 1 + 1$

$= 4 \geq 0$; $\hat{y} = 1$.

$w_1^{New} = 0 + 1(1 - 1) \times 7 = 0$

$w_2^{New} = 1 + 1(1 - 1) \times 3 = 1$.